

Urine Analyzer Lite — Overview

Setup, calibration, firmware, and operation guide

Compact, automated urine dipstick reader by **Levram Lifesciences**.

Single-strip, N20-motor-driven, ESP32-S3 based with thermal print output.

Overview

The Urine Analyzer Lite reads standard 10-pad reagent dipsticks using dual TCS34725 RGB color sensors and classifies results for 10 analytes. A compact N20 gear motor steps the strip under the optical head at controlled speed. Results print via an integrated CSN-A2 thermal printer.

Key specs:

Parameter	Value
MCU	ESP32-S3-DevKitC-1 (Xtensa dual-core 240 MHz, 8 MB flash)
Sensors	TCS34725 ×2, SHT31, BH1750, ADS1115, HX711
Display	Waveshare 4.3" MIPI DSI capacitive touch LCD
Motor	GA12-N20 6V 200RPM with Hall encoder + MX1508 H-bridge
Printer	CSN-A2 58mm thermal (ESC/POS)
Power	3.7V 2000mAh LiPo → TP4056 USB-C charger → S13V25F5 5V regulator
Sterilization	275nm UV-C LED (10s cycle post-scan)
Analytes	Leukocytes, Nitrites, Urobilinogen, Protein, pH, Blood, Specific Gravity, Ketones, Bilirubin, Glucose
Estimated BOM	~\$387 per unit (prototype); target <\$250 at 50-unit volume

Repository Structure

```
urine-analyzer-lite/  
  ■■■ hardware/  
    ■ ■■■ schematic/  
      ■ ■ ■■■ gen_schematic.py # KiCad schematic generator  
      ■ ■ ■■■ urine_analyzer_lite.kicad_sch  
      ■ ■■■ cad/  
        ■ ■■■ main_chassis.scad # OpenSCAD enclosure (all 3D-printed parts)  
    ■■■ firmware/  
      ■ ■■■ main/  
        ■ ■■■ main.ino # Arduino sketch (state machine)  
        ■ ■■■ config.h # Pin map and tunable constants  
        ■ ■■■ sensors.h # All sensor drivers  
        ■ ■■■ motor.h # N20 motion control + homing  
        ■ ■■■ analysis.h # HSV pad classification (10 analytes)
```

- ■■■ printer.h # ESC/POS thermal print
- ■■■ bom/
- ■■■ bom.csv # Full bill of materials (~40 items)
- ■■■ TODO.md # Project checklist
- ■■■ README.md

Hardware

Schematic

Open in KiCad 7+:

```
File → Open → hardware/schematic/urine_analyzer_lite.kicad_sch
```

To regenerate after edits:

```
cd hardware/schematic && python3 gen_schematic.py
```

CAD

Open in OpenSCAD ≥ 2021.01:

```
hardware/cad/main_chassis.scad
```

All 3D-printable parts are exported from one file. Render individual modules:

```
openscad -o chassis_base.stl main_chassis.scad
```

Print settings:

Part	Material	Infill	Layer
Chassis base, MCU tray, dipstick tray	PLA	20%	0.2 mm
Optical chamber, UV-C shield	Black PLA	100%	0.1 mm
Sensor mounts, printer bracket, belt parts	PETG	30%	0.2 mm

Firmware

Requirements

- Arduino IDE ≥ 2.3 with ESP32 board package ≥ 2.0.14
- Board: **ESP32S3 Dev Module** — Flash: 8MB, PSRAM: OPI, USB: Hardware CDC

Libraries (Arduino Library Manager)

```
Adafruit TCS34725 >= 1.4.2
Adafruit ADS1X15 >= 2.5.0
Adafruit SHT31 >= 2.2.2
BH1750 by Christopher Laws >= 1.3.0
HX711 by bogde >= 0.7.5
```

Flash

1. Open `firmware/main/main.ino` in Arduino IDE
2. Select board and port
3. Upload

Calibration (required before first use)

1. **White standard** — place Spectralon pad under optical head; record R/G/B values in `config.h` (`CAL_R/G/B_WHITE`)
2. **Scale** — place known weight on scale platform; adjust `scale.set_scale()` factor in `sensors.h`
3. **Motor steps** — measure real pad pitch with calipers; adjust `TICKS_PER_PAD` in `config.h`
4. **Hue ranges** — run known positive/negative strips; refine `PAD_TABLE[]` in `analysis.h`

Operation

1. Insert dipstick tray with strip face-up
2. Place sample cup on weight sensor (minimum 1 g detected)
3. Press scan button
4. Device homes, steps across all 10 pads, classifies, prints report
5. UV-C sterilization cycle runs automatically (10 s)
6. Remove strip; device returns to IDLE

Bill of Materials

See `bom/bom.csv` — estimated prototype total: **\$387 USD**.

License

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